BCS-062	GRAPH THEORY	L T P 3 0 0	3 Credits

Pre-requisites: None

Course Objectives:

CO1	Learn the fundamental concepts in graph theory in view of its applications in modern science.
CO2	Learn to understand and create mathematical proofs, including an appreciation of its significance in C S.
CO3	Use the concepts of Graph theory in subsequent courses in the design and analysis of algorithms, computability theory, software engineering and computer systems.
CO4	Apply concepts of the theory of probability in study of random phenomena, analyzing and interpreting data that involves uncertainties.
CO5	Learn the fundamental concepts in graph theory in view of its applications in modern science.

Detailed Syllabus

MODULE I

Graphs, Sub graphs, some basic properties, various example of graphs & their sub graphs, walks, path & circuits, connected graphs, disconnected graphs and component, euler graphs, various operation on graphs, Hamiltonian paths and circuits, the traveling sales man problem.

MODULE II

Trees and fundamental circuits, distance diameters, radius and pendent vertices, rooted and binary trees, on counting trees, spanning trees, fundamental circuits, finding all spanning trees of a graph anda weighted graph, algorithms of primes, Kruskal and Dijkstra Algorithms. Cuts sets and cut vertices, some properties, all cut sets in a graph, fundamental circuits and cut sets , connectivity and separability, network flows Planer graphs, combinatorial and geometric dual: Kuratowski graphs, detection of planarity, geometric dual, Discussion on criterion of planarity, thickness and crossings.

MODULE III

Vector space of a graph and vectors, basis vector, cut set vector, circuit vector, circuit and cut set subspaces, Matrix representation of graph – Basic concepts; Incidence matrix, Circuit matrix, Path matrix, Cut-set matrix and Adjacency matrix. Coloring, covering and partitioning of a graph, chromatic number, chromatic partitioning, chromatic polynomials, matching, covering, four color problem Discussion of Graph theoretic algorithm wherever required.

Text Books:

1. Deo, N, Graph theory with applications to Engineering and Computer Science, PHI
2. Gary Chartrand and Ping Zhang, Introduction to Graph Theory, TMH
3. Robin J. Wilson, Introduction to Graph Theory, Pearson Education
4. Harary, F, Graph Theory, Narosa
5. Bondy and Murthy: Graph theory and application. Addison Wesley.
6. V. Balakrishnan, Schaum's Outline of Graph Theory, TMH
7. Geir Agnarsson, Graph Theory: Modeling, Applications and Algorithms, Pearson Education

Course Outcomes: After the completion of the course the student will be able to:

CO1	Able to define the basic concepts of graphs, directed graphs, and weighted graphs.
CO2	Able to define the properties of bipartite graphs, particularly in trees.
CO3	Able to understand the concept of colourings and theory.
CO4	Able to understand the concept of Trees and fundamental circuits
CO5	Able to understand the concept of Kruskal and Dijkstra Algorithms